

What is Caching in Snowflake?

Snowflake provides **high performance query execution** using its **Multi-Cluster Shared Data Architecture** and **multiple caching layers**.

Caching allows Snowflake to:

- Return query results faster
- Reduce data scanning from storage
- Save compute cost and credits

If required data is already available in cache, Snowflake **does not read data from remote storage again**.

This significantly improves **query performance**.

Types of Snowflake Caches

Snowflake has **three main caching layers**:

- 1 Metadata Cache
- 2 Result Cache
- 3 Virtual Warehouse Cache (SSD Cache)

Snowflake Query Optimizer checks caches in this order:

Result Cache → Warehouse Cache → Remote Storage

Only if data is not found in cache does Snowflake read from **remote storage**.

1 Metadata Cache

Metadata Cache stores **information about database objects and micro-partitions**.

This metadata is maintained in the **Snowflake Cloud Services Layer**.

Important point:

Metadata queries **do not require a virtual warehouse**.

Information Stored in Metadata Cache

Snowflake stores the following metadata:

- Row count of tables
- Table size in bytes
- File references
- Table versions
- Micro-partition metadata
- Clustering information

- Column statistics
- Minimum values (MIN)
- Maximum values (MAX)
- Number of distinct values

This metadata helps Snowflake perform **partition pruning**.

Example

If a query filters:

```
SELECT *  
FROM orders  
WHERE order_date = '2025-01-01';
```

Snowflake checks **metadata statistics** to identify which **micro-partitions contain this date**.

It scans only those partitions and skips the rest.

This reduces:

- Data scanning
 - Query execution time
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2 Result Cache

Result Cache stores the **final query results**.

If the **exact same query** is executed again, Snowflake returns the result **directly from the cache**.

No compute resources are used.

Result cache retention:

24 hours by default

If the query is re-executed within 24 hours, the timer resets and results can be reused **up to 30 days**.

Important Conditions for Result Cache

Result Cache works only if:

- Query text is exactly the same
- Underlying data has not changed
- Role is the same
- Query does not contain non-deterministic functions

Example of functions that invalidate cache:

```
CURRENT_TIME()  
CURRENT_TIMESTAMP()
```

Result Cache Characteristics

- Shared across Virtual Warehouses
- Different users can reuse cache
- Role must be the same
- No compute credits are consumed

Example scenario:

User A runs query at 9 AM.

User B runs the **same query** later.

Snowflake returns results **instantly using result cache**.

3 Virtual Warehouse Cache (SSD Cache)

Every Virtual Warehouse has its own **local SSD storage cache**.

This cache stores **raw data from micro-partitions** when a query is executed.

When Snowflake reads data from remote storage, it stores that data in the **warehouse SSD cache**.

If the same data is queried again:

Snowflake retrieves it from the **warehouse cache instead of remote storage**.

Key Characteristics

- Each warehouse has its own cache
 - Cache stores raw micro-partition data
 - Cache exists only while warehouse is running
 - Cache is cleared when warehouse is suspended
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Example

First query execution:

Data is retrieved from **remote storage**.

Second execution:

Data is retrieved from **warehouse SSD cache**.

This reduces query latency.

Cache Data Categories

Snowflake categorizes data based on where it is stored.

Layer	Data Type
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Result Cache	Hot Data
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Warehouse Cache	Warm Data
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Remote Storage	Cold Data
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Hot data is fastest to access.

Query Execution Flow in Snowflake

When a query runs, Snowflake checks data in this order:

- 1 Result Cache
- 2 Warehouse SSD Cache
- 3 Remote Storage

If data exists in **result cache**, no compute is used.

If data exists in **warehouse cache**, warehouse compute scans cached data.

If data does not exist in any cache, Snowflake reads from **remote storage**.

Query Execution Example

First query execution:

Data comes from **Remote Disk**.

Bytes scanned → Green

Second execution of same query:

Data comes from **Result Cache**.

No compute used.

If query text slightly changes:

Example:

SELECT amount FROM orders

vs

SELECT amount AS order_amount FROM orders

Result cache will not be used.

Snowflake will retrieve data from **Warehouse Cache**.

Bytes scanned → Blue

Cache Expiration

Result Cache expires if:

- 24 hours pass without query execution
- Underlying table data changes

However, if the query is executed within 24 hours repeatedly, cache can extend up to **30 days**.

Example timeline:

Query run Monday 9 AM

Run again Tuesday 8 AM → cache refreshed

Run again Wednesday → extended

Cache Behavior When Warehouse Suspends

If warehouse is suspended:

- SSD cache is cleared
- Warehouse cache becomes empty

When warehouse resumes:

Data must be read again from **remote storage**.

Benefits of Snowflake Caching

- Faster query performance
- Reduced data scanning
- Lower compute cost
- Improved dashboard performance
- Efficient repeated query execution

Disabling Caches in Snowflake

Sometimes caches are disabled when testing **real query performance**.

Disable Result Cache

```
ALTER SESSION SET USE_CACHED_RESULT = FALSE;
```

This forces Snowflake to execute the query again instead of using cached results.

Clear Warehouse Cache

Warehouse cache can be cleared by suspending the warehouse.

```
ALTER WAREHOUSE COMPUTE_WH SUSPEND;
```

When warehouse resumes, cache will be empty.

Metadata Cache

Metadata cache **cannot be disabled** because it is part of the Snowflake **Cloud Services layer**.

Important Interview Points

Snowflake caching layers:

- Metadata Cache
- Result Cache
- Warehouse Cache

Result cache validity:

24 hours (extendable up to 30 days)

Warehouse cache cleared when:

Warehouse is suspended

Metadata cache:

Does not require warehouse

Quick Summary

Snowflake caching improves performance by storing **query results and data in multiple cache layers**.

Instead of repeatedly scanning remote storage, Snowflake retrieves data from cache whenever possible.

This reduces **query latency, compute usage, and overall cost**.